

Roll No. ....

**DETF-03**

**B. TECH. (FIFTH SEMESTER)**  
**MID SEMESTER EXAMINATION, Oct., 2022**  
**REFRIGERATION AND AIR CONDITIONING**

**Time : 1½ Hours**

**Maximum Marks : 50**

**Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain the different methods of cooling with examples. (CO1, CO2)

OR

- (b) Explain reversed Carnot cycle with the help of T-s diagram. Also, explain why it is not a practical cycle. (CO2, CO3)

2. (a) A cold storage plant is required to store 20 tonnes of fish. The fish is supplied at a temperature of 30°C. The specific heat of fish above freezing point is 2.93 kJ/kg K. The specific heat of fish below freezing point is 1.26 kJ/kg K. The fish is stored in cold storage which is maintained at - 8°C. The freezing point of fish is - 4°C. The latent heat of fish is 235 kJ/kg. If the plant requires 75 kW to drive it. (CO6)

Find :

- (i) The capacity of the plant, and (ii) Time taken to achieve cooling.

Assume actual C. O. P. of the plant as 0.3 of the Carnot C. O. P.

*P. T. O.*

OR

- (b) An ice plant produces  $10 \times 10^3$  kg of ice per day at  $0^\circ\text{C}$  using water at a temperature of  $23^\circ\text{C}$ . Find the amount of heat transferred from the system per minute. Also, estimate the power required by the compressor motor, if the COP of the plants is 3.5. (CO6)
3. (a) Write the advantages and disadvantages of air refrigeration system. Also explain the working of Bootstrap air refrigeration system. (CO2, CO4)

OR

- (b) Explain the working of Regenerative air refrigeration system with the help of a neat diagram and also draw this cycle on T-s diagram. (CO2, CO4)
4. (a) Explain various factors affecting the performance of VCRC. (CO4)

OR

- (b) A refrigerator used R-12 as a working fluid and it operates on an ideal vapour compression cycle. The temperature of refrigerant in the evaporator is  $-20^\circ\text{C}$  and in the condenser is  $40^\circ\text{C}$ . The refrigerant is circulated at the rate of 0.03 kg/s. Determine the coefficient of performance and capacity of refrigeration plant in the TR. (CO6)

Use the following data for R-12 :

$$\text{At } -20^\circ\text{C} \quad h_1 = 178.61 \text{ kJ/kg,}$$

$$s_1 = 0.7082 \text{ kJ/kg}$$

$$\text{At } 40^\circ\text{C} \quad h_3 = h_{f_3} = 74.53 \text{ kJ/kg}$$

5. (a) A vapour compression refrigerator works between the pressure limits of 60 bar and 25 bar. The working fluid is just dry at the end of compression and there is no under cooling of the liquid before the expansion valve. (CO6)

(3)

Determine :

- (i) C. O. P. of the cycle, and
- (ii) Capacity of the refrigerator if the fluid flow is at the rate of 5 kg/min.

| Pressure<br>(bar) | Saturation<br>temperature<br>(k) | Enthalpy (kJ/kg) |        | Entropy (kJ/kg K) |        |
|-------------------|----------------------------------|------------------|--------|-------------------|--------|
|                   |                                  | Liquid           | Vapour | Liquid            | Vapour |
| 60                | 295                              | 151.96           | 293.29 | 0.554             | 1.0332 |
| 25                | 261                              | 56.32            | 322.58 | 0.226             | 1.2464 |

OR

- (b) What do you mean by Cryogenics ? Explain Cascade refrigeration systems with a neat diagram. (CO<sub>2</sub>, CO<sub>3</sub>)